

Moradabad Institute of Technology
Computer Science & Engineering and Allied

Class Test-1

Session: 2025-26

Subject Name: Data Structure
Subject Code: BCS301
Branch: CSE, AIML, DS, IOT
Year: 2nd

MM: 20

Duration: 1hr

Section: A, B, C, D, I, J

Sem: 3rd

Q. No	1	2	3	4	5	6
CO	CO1	CO2	CO1	CO1	CO2	CO3
Bloom's Level	K2	K3	K3	K3	K3	K3

CO1	Implement and explain how arrays and linked lists are represented in memory, used by the algorithms and their common applications.
CO2	Implement and use linear data structures stacks and queues in computer science applications.
CO3	Implement the concept of recursion, application of recursion and removal of recursion.

Note:

This paper contains two sections (A) and (B). All sections are compulsory.

21
12/6

Section (A)		Marks
Attempt all questions.		
1)	Explain Linear and Non-Linear Data Structure with example.	(2)
2)	Explain stack. Write C function for push and pop operations. Assume stack is stored in array.	(3)
3)	Write a C function to insert an element at the end of a doubly linked list.	(3)
Section (B)		
Attempt all questions.		
4)	Consider the following multidimensional array, X [-2:3, 1:3, 0:6]. a). Find the length of each dimension b). Find number of elements in X. c). Find the address of element X [3, 2, 4] if X is stored in row major. Assume Base(X) = 200, w = 4.	(4)
5)	Write an algorithm to convert infix expression to postfix. Convert the following infix expression to Postfix form using stack: A + (B*C/E*F + G^H*I) + J*K	(4)
6)	Explain recursion. Write a recursive function to find the factorial of any number. Explain the limitations of recursion?	(4)

MORADABAD INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
SESSIONAL TEST 2
Set 1

Course: B.Tech
 SESSION: 2025-26
 Subject: Data Structure
 Max Marks: 15

Semester: 3rd
 Section: A, B, C, D, I, J
 Subject Code: BCS301
 Time: 1HR

Q. No	1	2	3	4	5	6
CO	CO5	CO5	CO4	CO4	CO5	CO5
Bloom's Level	K3	K3	K4	K3	K4	K3

CO4	Arrange the data using various internal and external sorting techniques, Implement and analyze searching and sorting techniques.
CO5	Implement non-linear data structures tree and graph.

Attempt all questions	Marks
Section A	
1. Construct the Binary Tree from the inorder and pre-order traversal given below and find the postorder traversal of the resulting binary tree. In-order: H, D, I, B, J, E, K, A, F, C, G Pre-order: A, B, D, H, I, E, J, K, C, F, G	(2)
2. Explain all the steps used to build Huffman Tree where following characters are given with their frequencies: a:5, b:9, c:12, d:13, e:16, f:45 Find the Huffman code for each character.	(3)
3. Write a c function (or algorithm) of merge-sort. Write the complexity of merge-sort. Sort the following elements using merge-sort: 25,12,5,18,7,3,1	(3)
Section B	
4. The keys 35, 15, 14, and 95 are inserted into an initially empty hash table of size 11 using open addressing with hash function $h(k) = k \bmod 10$ and linear probing. Find the resultant hash table?	(4)
5. Define AVL search Tree. Insert 14, 17, 11, 7, 53, 4, 13 into an empty AVL tree.	(4)
6. Find the Shortest path distance using Dijkstra algorithm from source vertex '5' to all vertices in the following graph. Also draw the shortest path tree.	(4)

